

Section A

1 (short)

- (a) Derive the Norton equivalent circuit of the circuit shown in Fig. 1 across terminals AB. [4]
- (b) For the case $R = 10\ \Omega$ and $I_0 = 3\text{ A}$, derive the corresponding Thévenin equivalent circuit. [3]
- (c) Consider the case where the current source has an internal resistance of $10\text{ k}\Omega$. Determine the power dissipated in the entire system. [3]

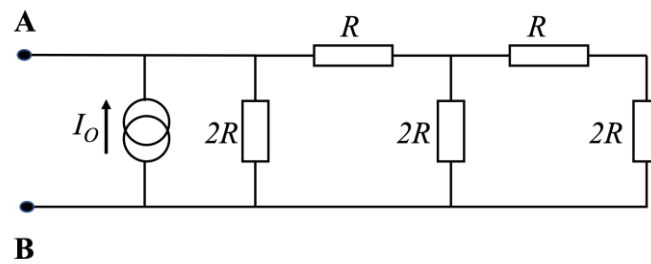


Fig. 1